

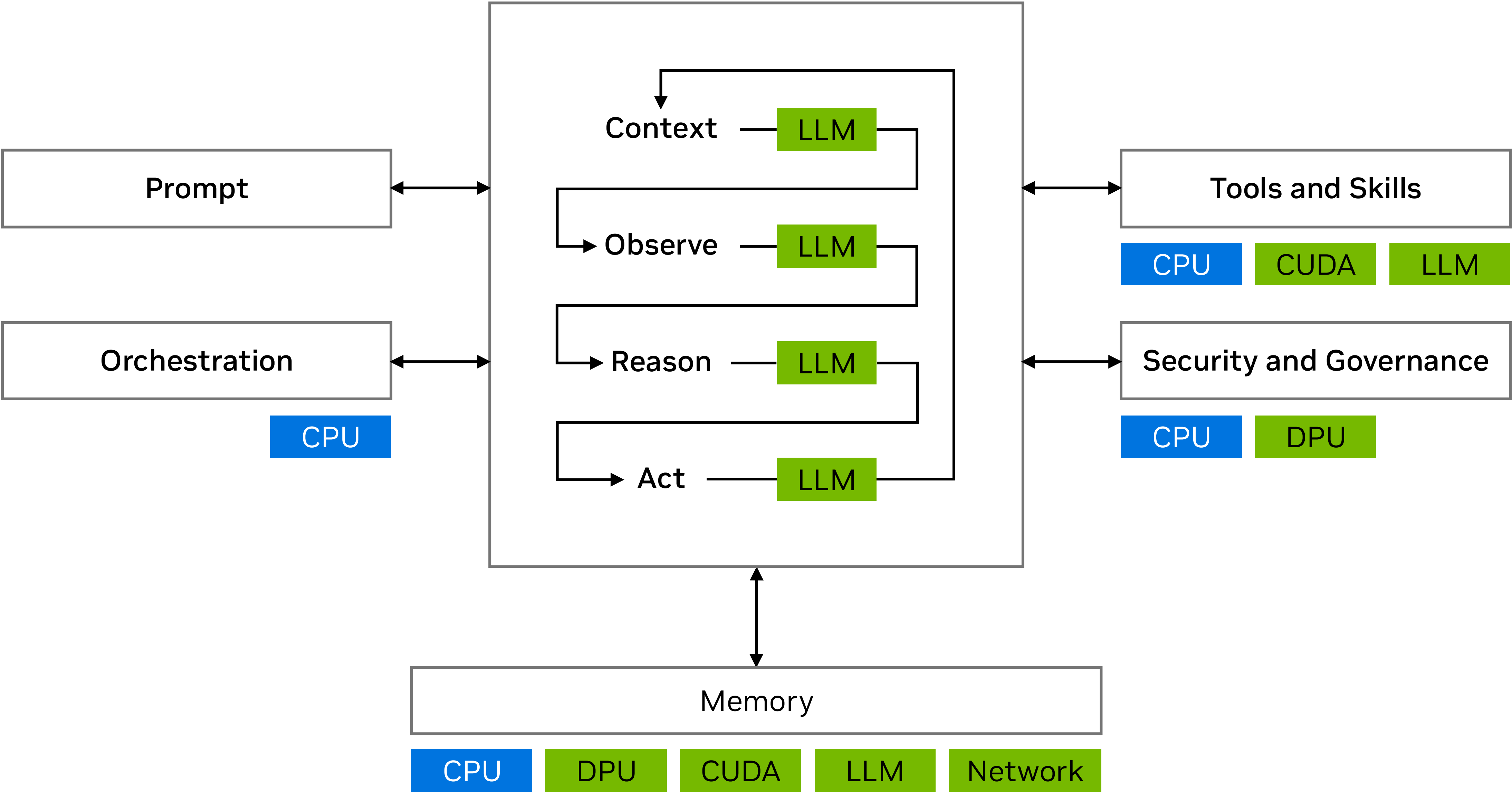


NVIDIA BlueField-4 Processor Powers the AI Factory

Idan Burstein
Hot Chips 2026



Agentic AI Is the Most Complex Computing Workload in History



NVIDIA Vera Rubin Is a Full-Stack AI Factory Platform

Extreme co-design for agentic AI across seven chips and five racks

Vera Rubin
NVL72



Foundation Platform

Groq 3
LPX



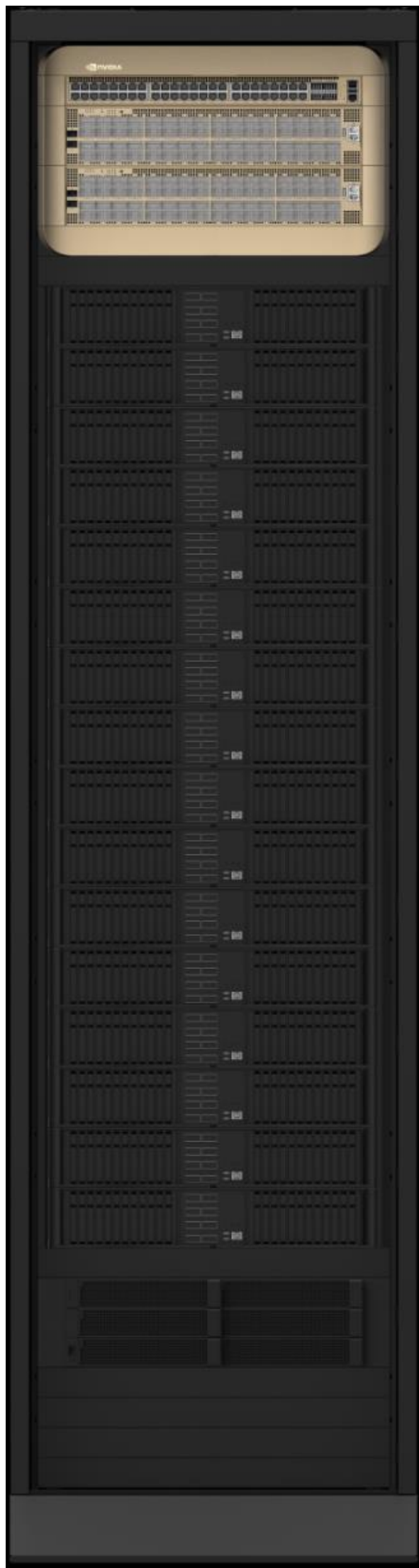
Extend Interactivity

Vera CPU
Rack



Tool Calls and Sandboxes

Vera BlueField-4
STX Storage



Context Memory Storage

Spectrum-6 SPX
Networking



Scale-Out Fabric

DPU Is the Processor for Networking, Security and Storage

Running data center operating system, required in every node

CPU

General-purpose compute

- Applications
- Operating systems
- Business logic

GPU

Accelerated compute

- AI / ML
- High-performance computing
- Parallel workloads

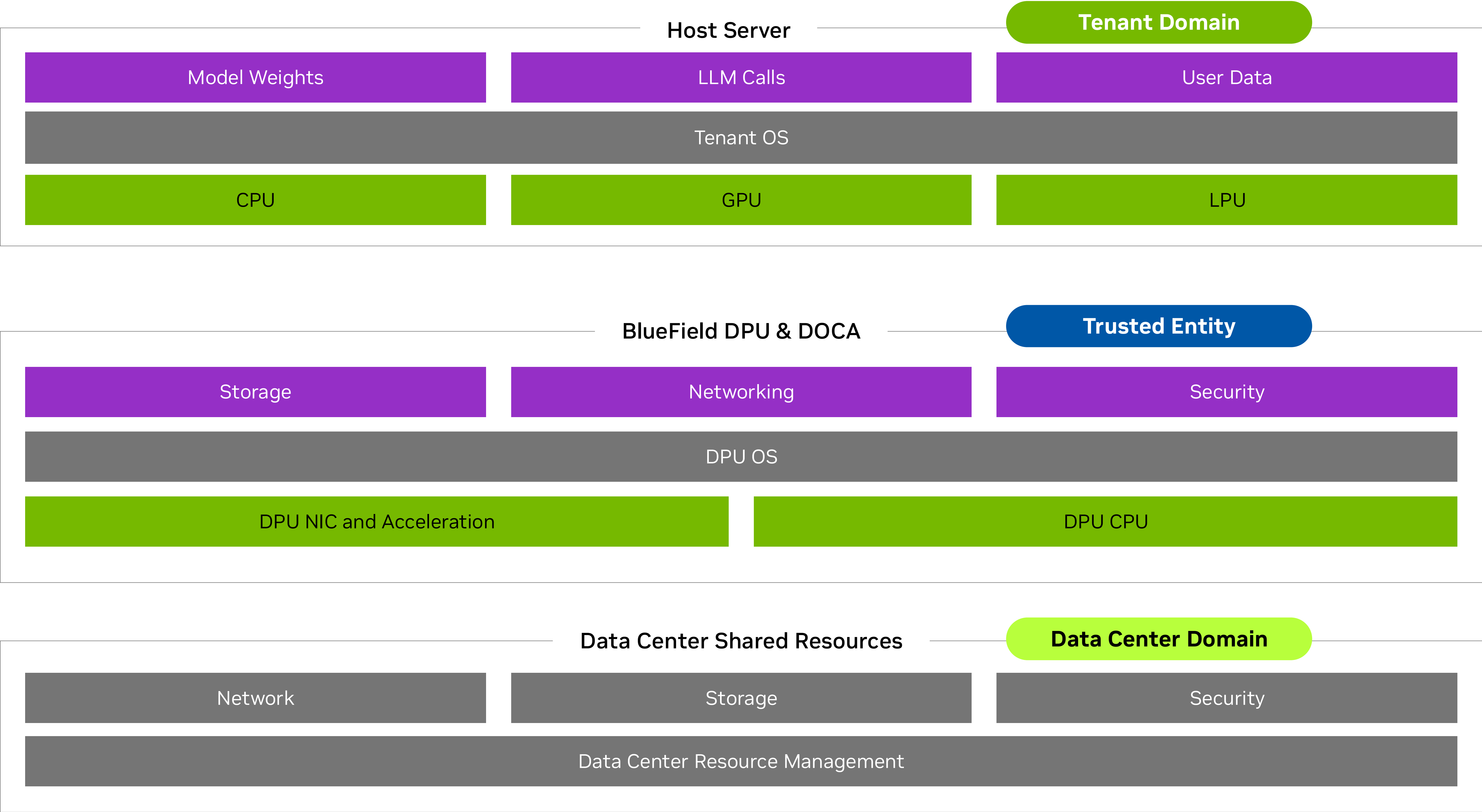
DPU

Data center processing

- Networking
- Storage
- Security

DPU is the Trusted Entity of the Platform

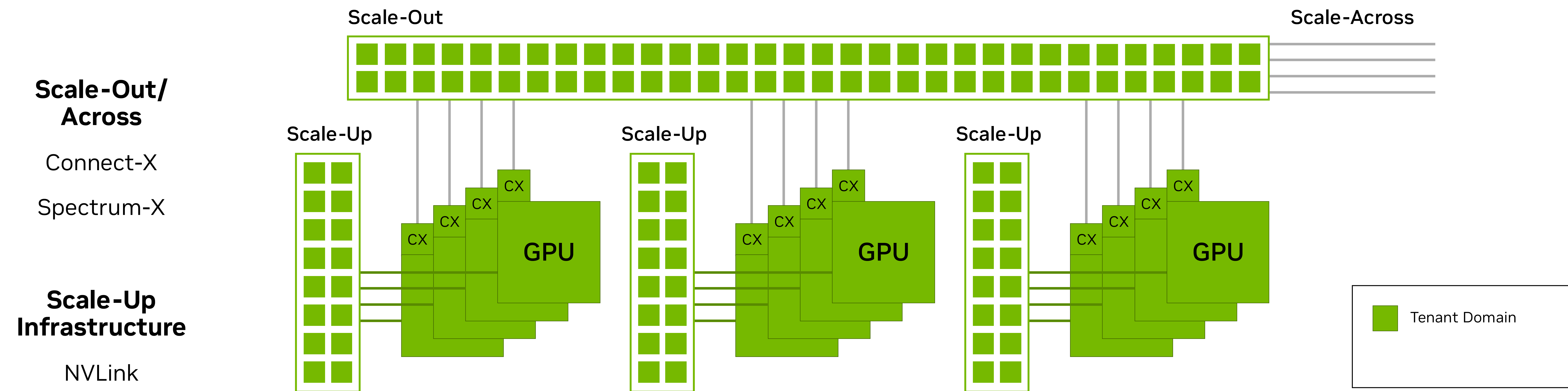
The DPU is a server in front of the server — a new trust domain isolated from the host and the data center resources



Scale-Up Makes Multiple GPUs Operate as One System

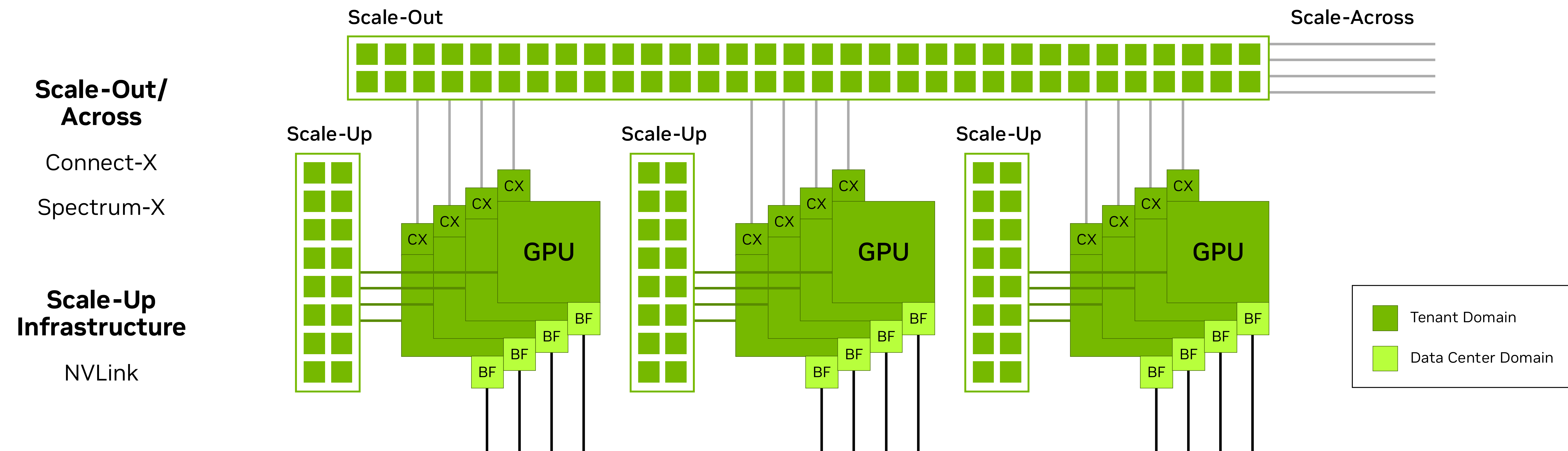


Scale-Out Transforms the Data Center to One Unit of Compute



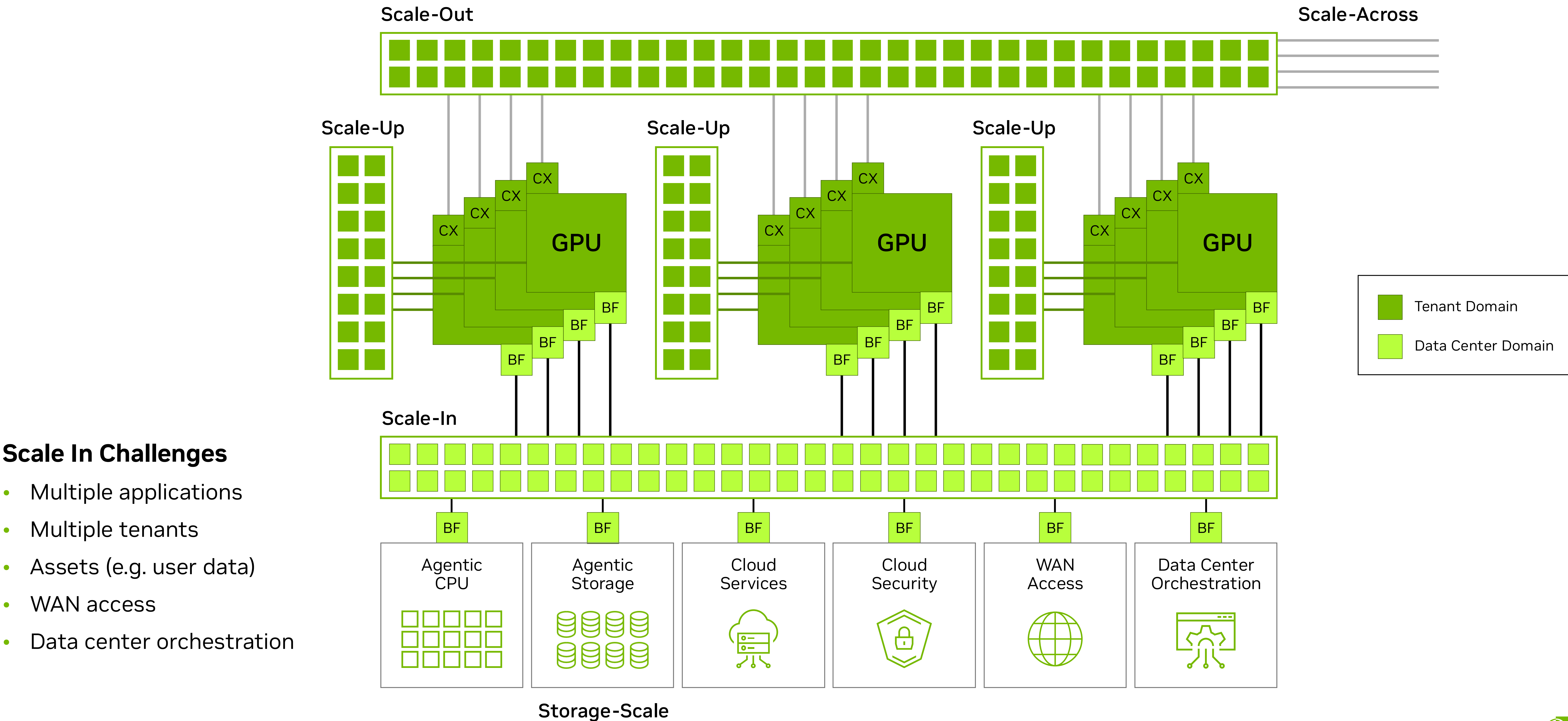
BlueField Is Co-Designed Into Every Vera Rubin System

Co-designed security, data, and infrastructure services



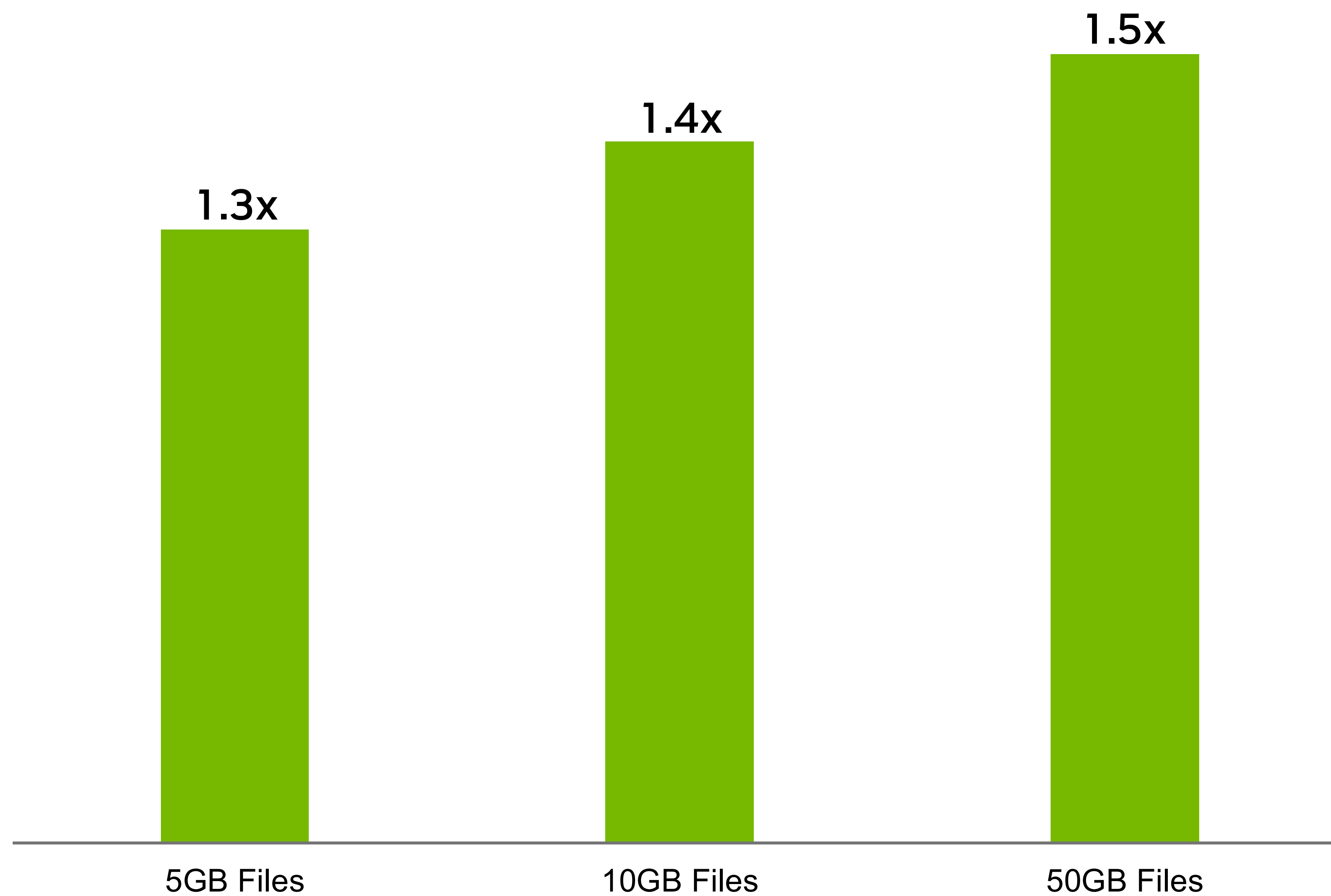
Announcing BlueField-4 Spectrum-X Scale-In Network

End to end data center networking solution designed for the diverse agentic workload



1.5x Faster Storage Access With NVIDIA Spectrum-X Ethernet

Spectrum-X Ethernet vs. OTS Ethernet

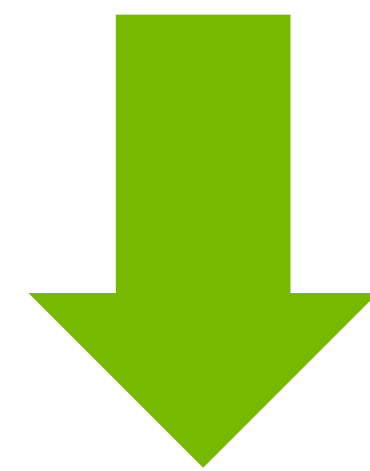


DPU Requires World-Class Network Acceleration and CPU

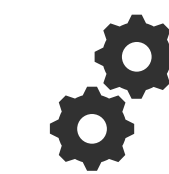


Network & data acceleration

Data transfers with RDMA and TCP/IP
Encryption & authentication
Firewall packet processing
Line-rate switching & routing

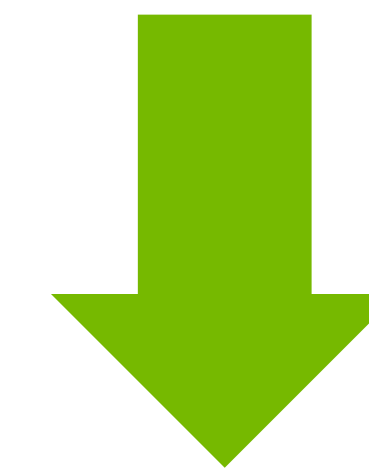


NVIDIA ConnectX-9



Complex control at high rate

Keys & session management
Firewall connection management
Storage volume & I/O management
Telemetry & health



NVIDIA Grace CPU

Cloud DPUs Were Built for General-Purpose Computing

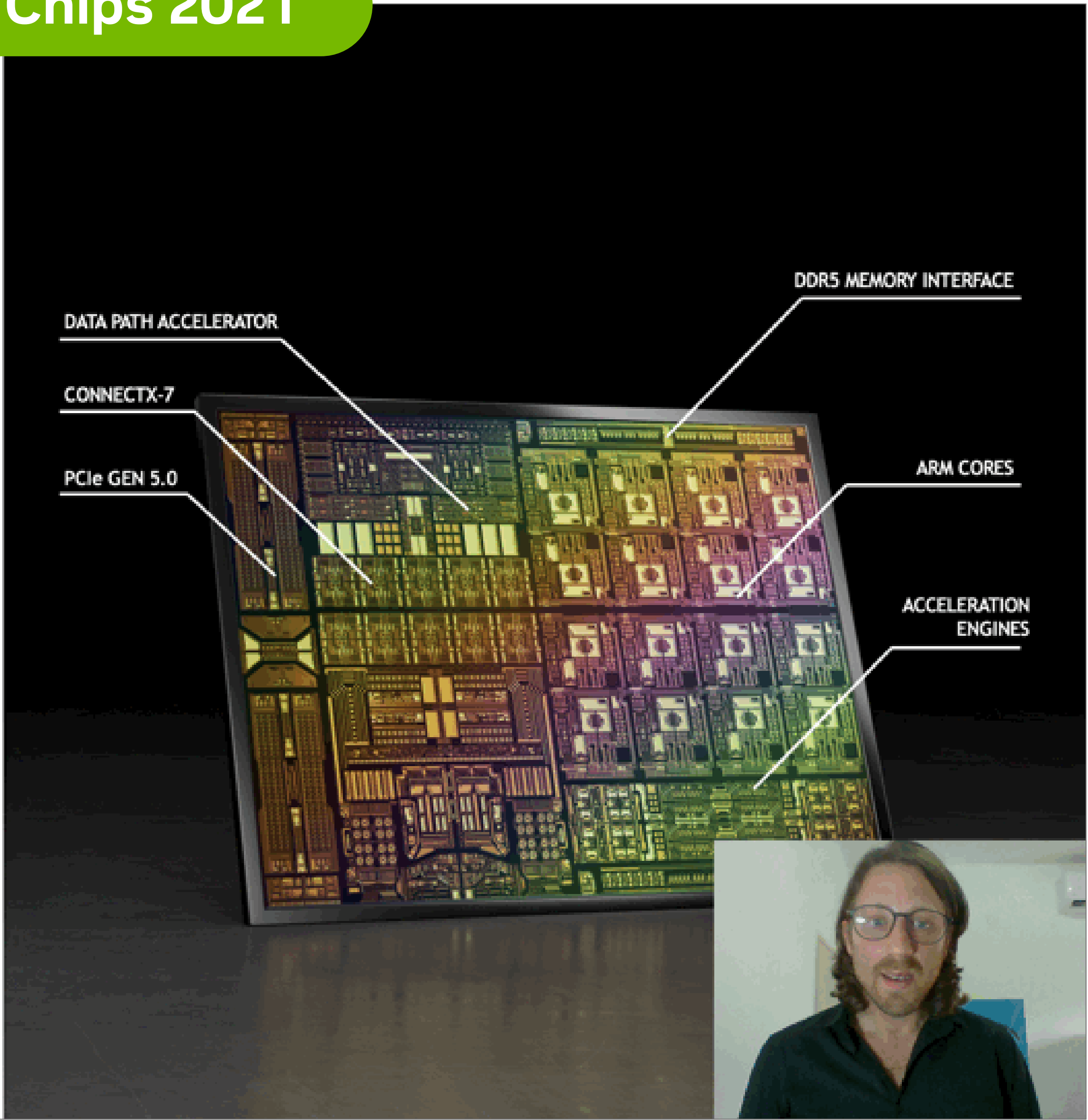
One size fits all DPU

Hot Chips 2021

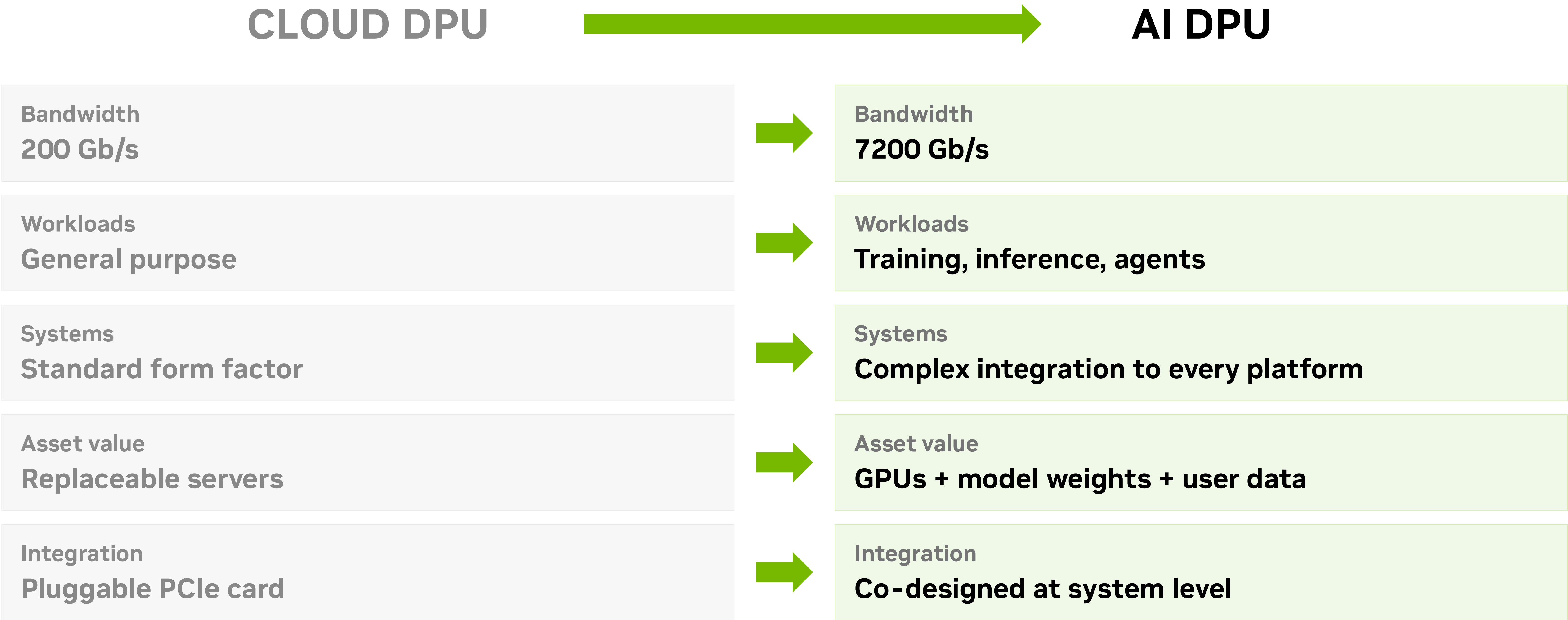
NVIDIA BLUEFIELD-3 DPU

First 400Gb/s Data Processing Unit

- 22 Billion Transistors
- 400Gb/s Ethernet & InfiniBand Connectivity (1-4 Ports)
- PCIe Switch Gen 3/4/5 x32+x4
- 400Gb/s Crypto / Security Acceleration
- 2x370M PPS, 2x40M PPS at scale of millions of flows
- 18M IOP/s Elastic Block Storage
- 300 Equivalent x86 Cores
- 16C A78 ARM 42 SPECINT2k17-rate
- 128b DDR5-5600

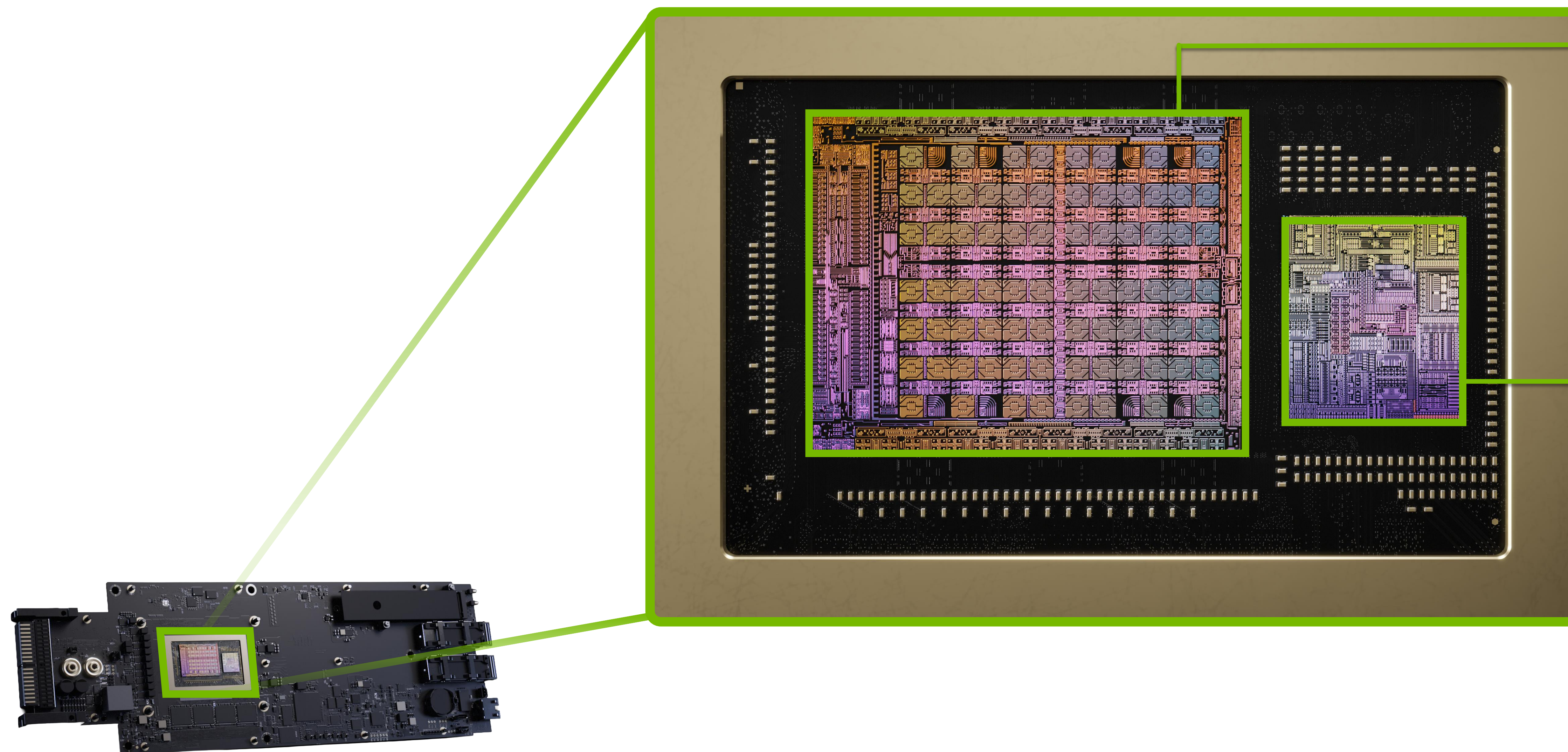


Agentic AI Requires a New Class of AI-Native Infrastructure



BlueField-4: The AI-Native Scale-In Infrastructure Engine

Integrating world class CPU and world class networking, storage and security acceleration



Grace CPU

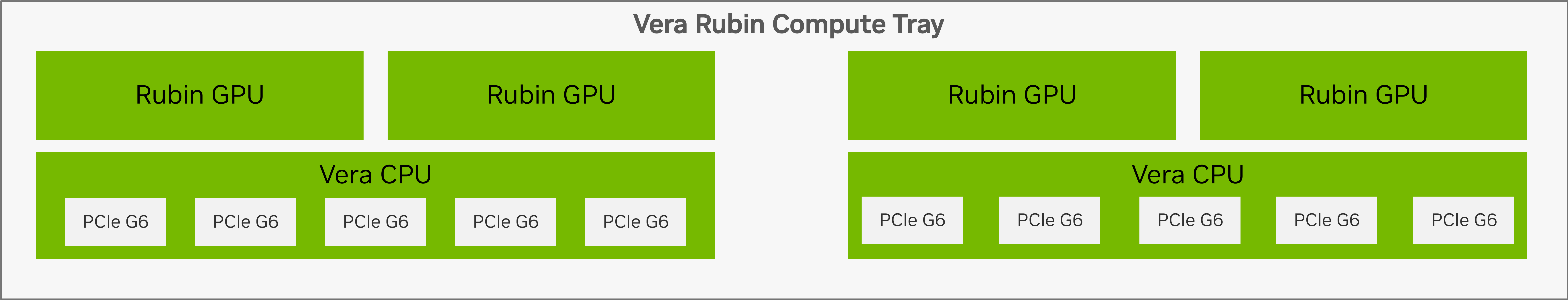
- CPU SPECINT 220 (6x BF3)
- CPU LPDDR5 Bandwidth 275GB/s
- 64 Arm Neoverse V2™ @1.7GHz
- PCIe and Astra

ConnectX-9 Networking

- 800G Ethernet (2x BF3)
- 200G PAM4 SerDes
- PCIe Gen6 x16
- Spectrum-X RDMA, MRC, RoCE, PRDMA
- Inline encryption (PSP, TLS, IPSEC, AES-XTS)
- 200 MPPS
- 25 MIOPs

Vera Rubin Server Extreme Bandwidth Requirements

Every compute tray needs 7Tb/s of aggregate network bandwidth



4x1.6 Tb/s

GPU scale-out

800 Gb/s

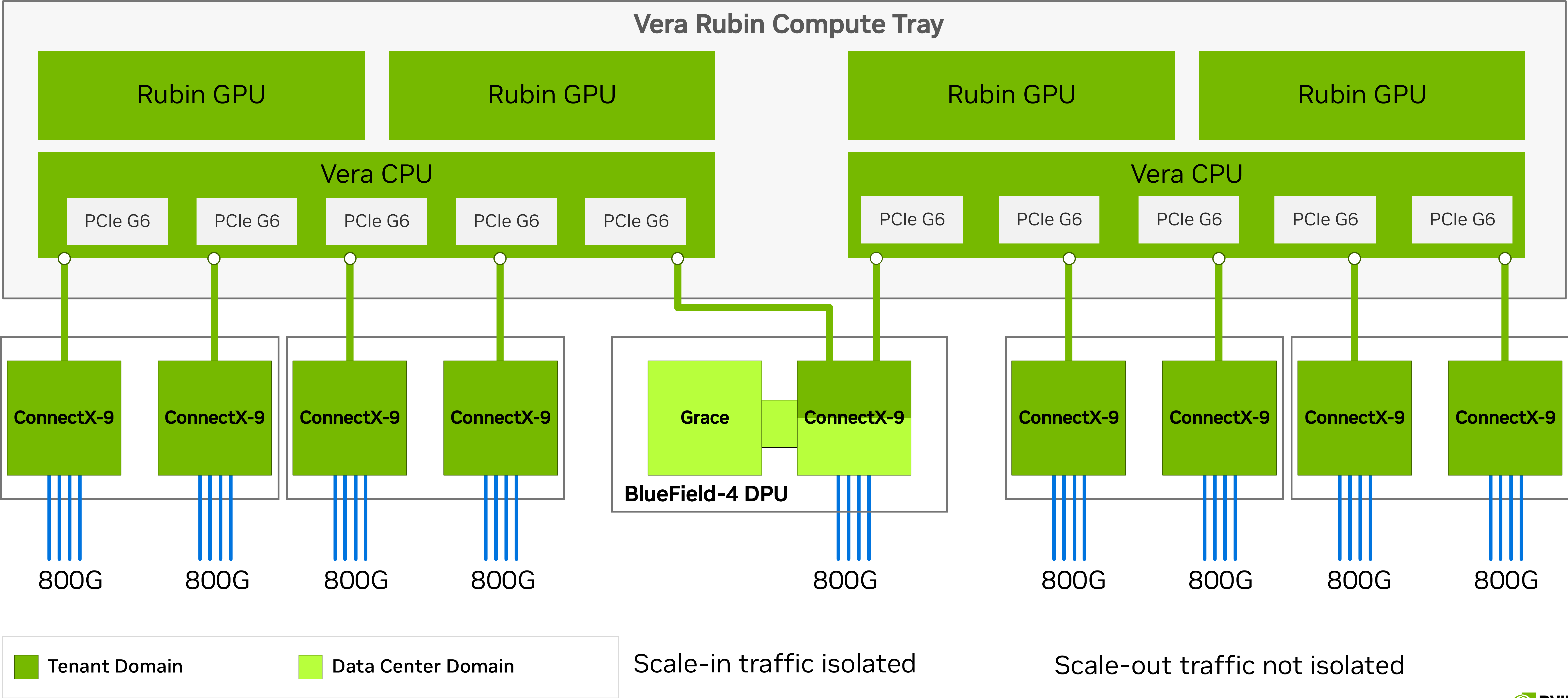
GPU scale-in

7200 Gb/s

Aggregate per Vera Rubin compute tray

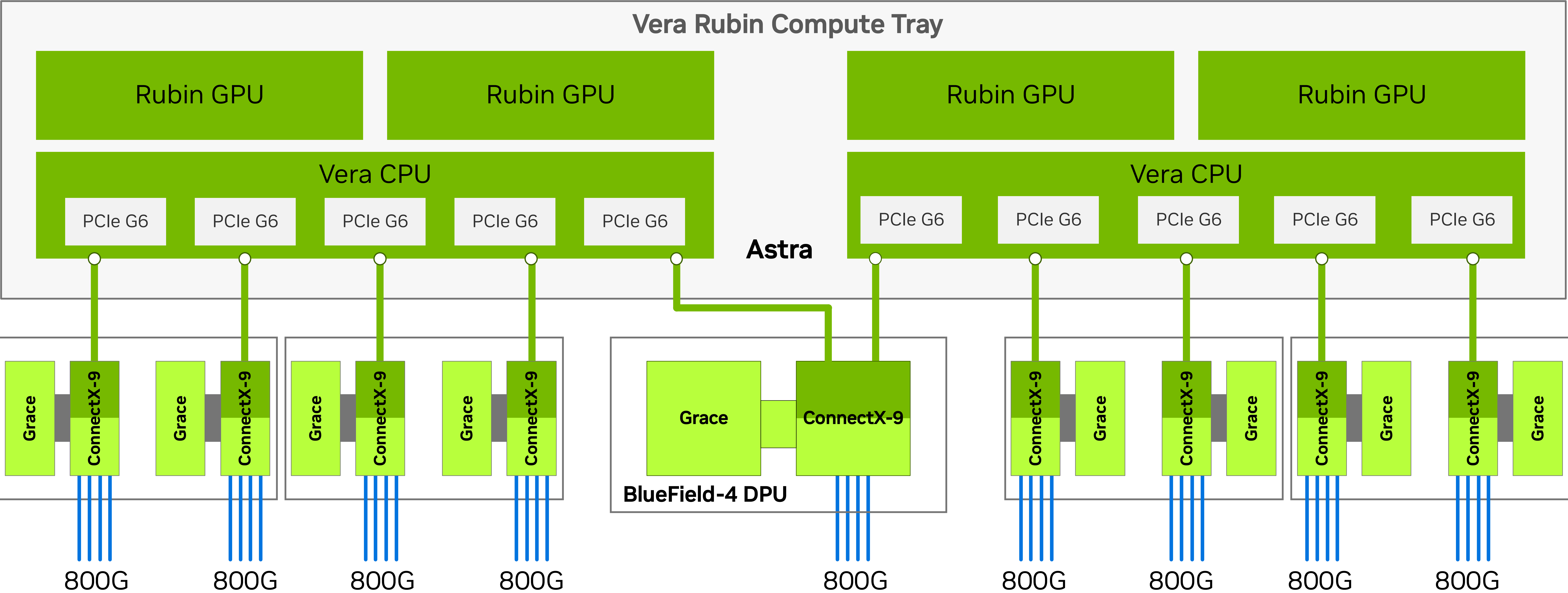
Vera Rubin: The Traditional Approach #1 - Reduced Security

GPU network can't be secured end to end isolated from the tenant



Vera Rubin: The Traditional Approach #2 - 4x Higher Power

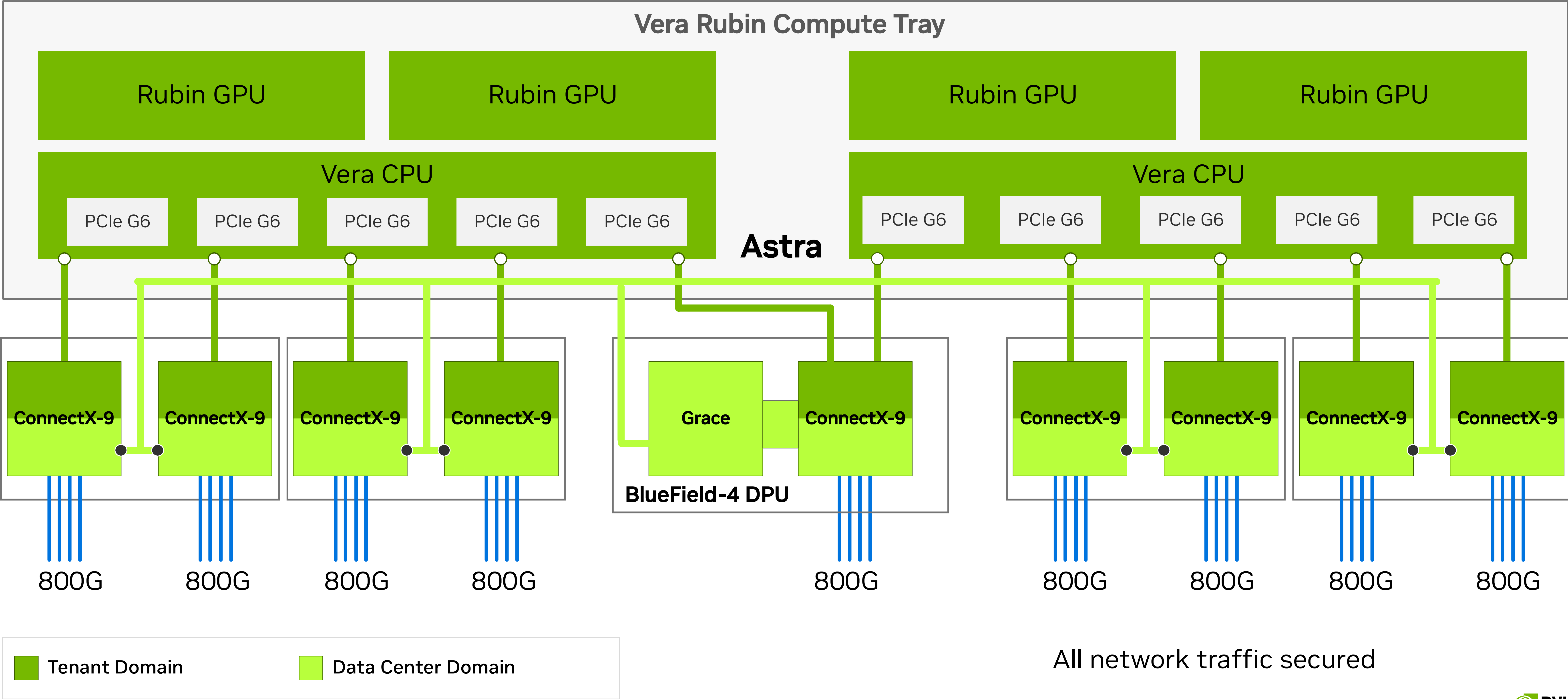
Every DPU needs—CPUs, memory system, connection to data center network, connection to management



Power, cost and connectivity complexity for every NIC

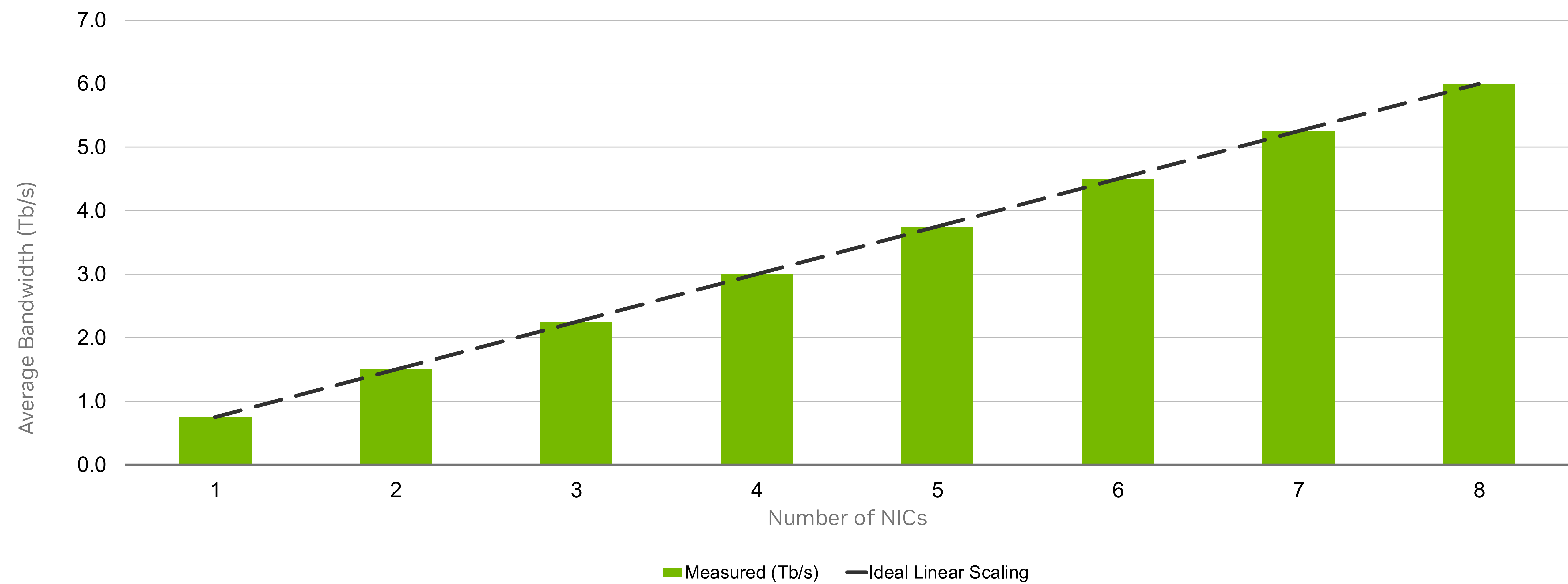
Power, cost and connectivity complexity for every NIC

Vera Rubin: BlueField-4 Astra Delivers 7200 Gb/s with CSP-Grade Security and Management

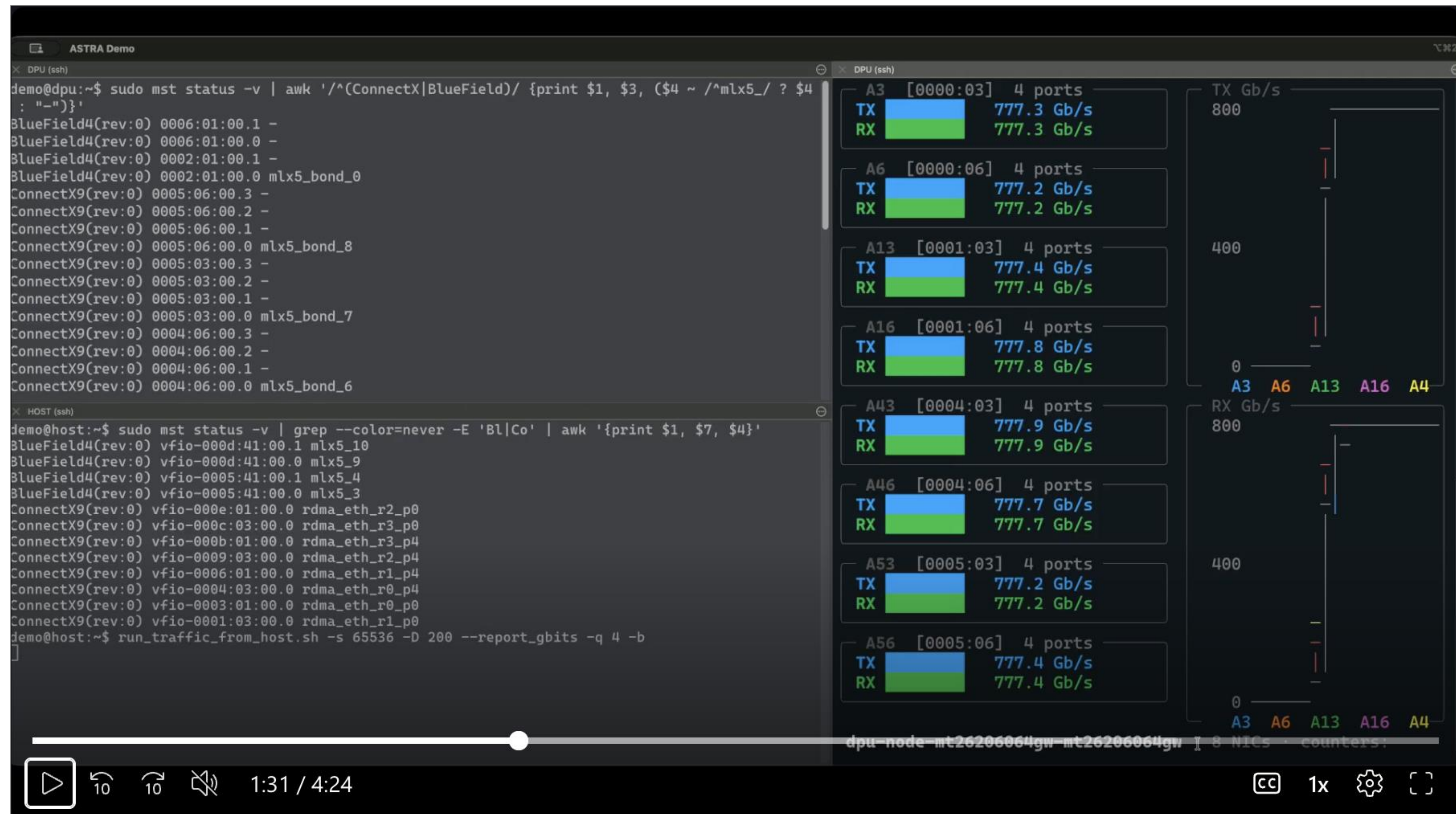


Scale Linearly to Multi-Terabit Networking Performance

Encrypted and virtualized scale out traffic secured and monitored by BlueField-4



Vera Rubin: BlueField-4 7Tb/s Platform Level DPU Live

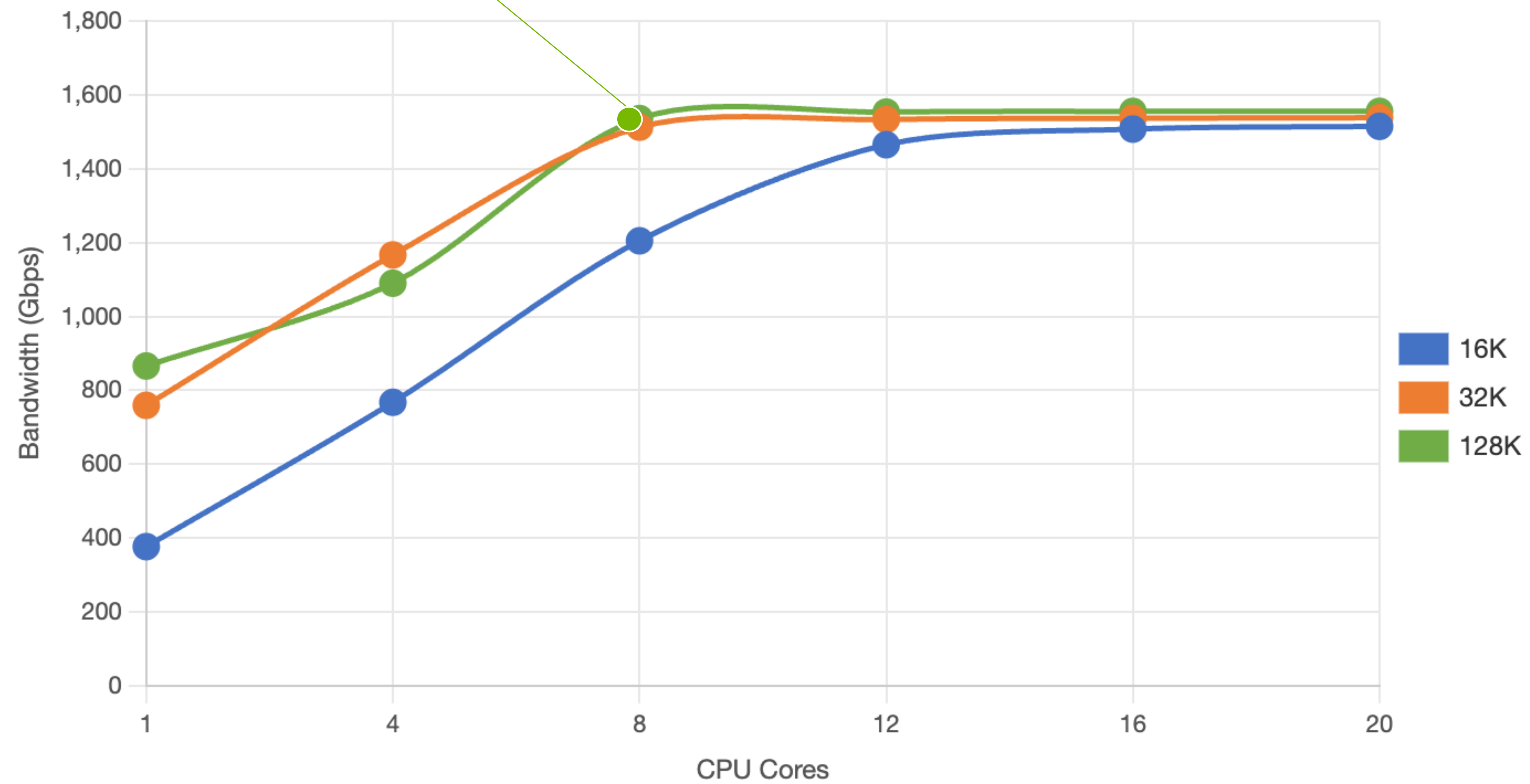


Unlock 2x Faster Data Access for Rubin GPUs

NVMe over fabrics performance running on BlueField-4 Grace

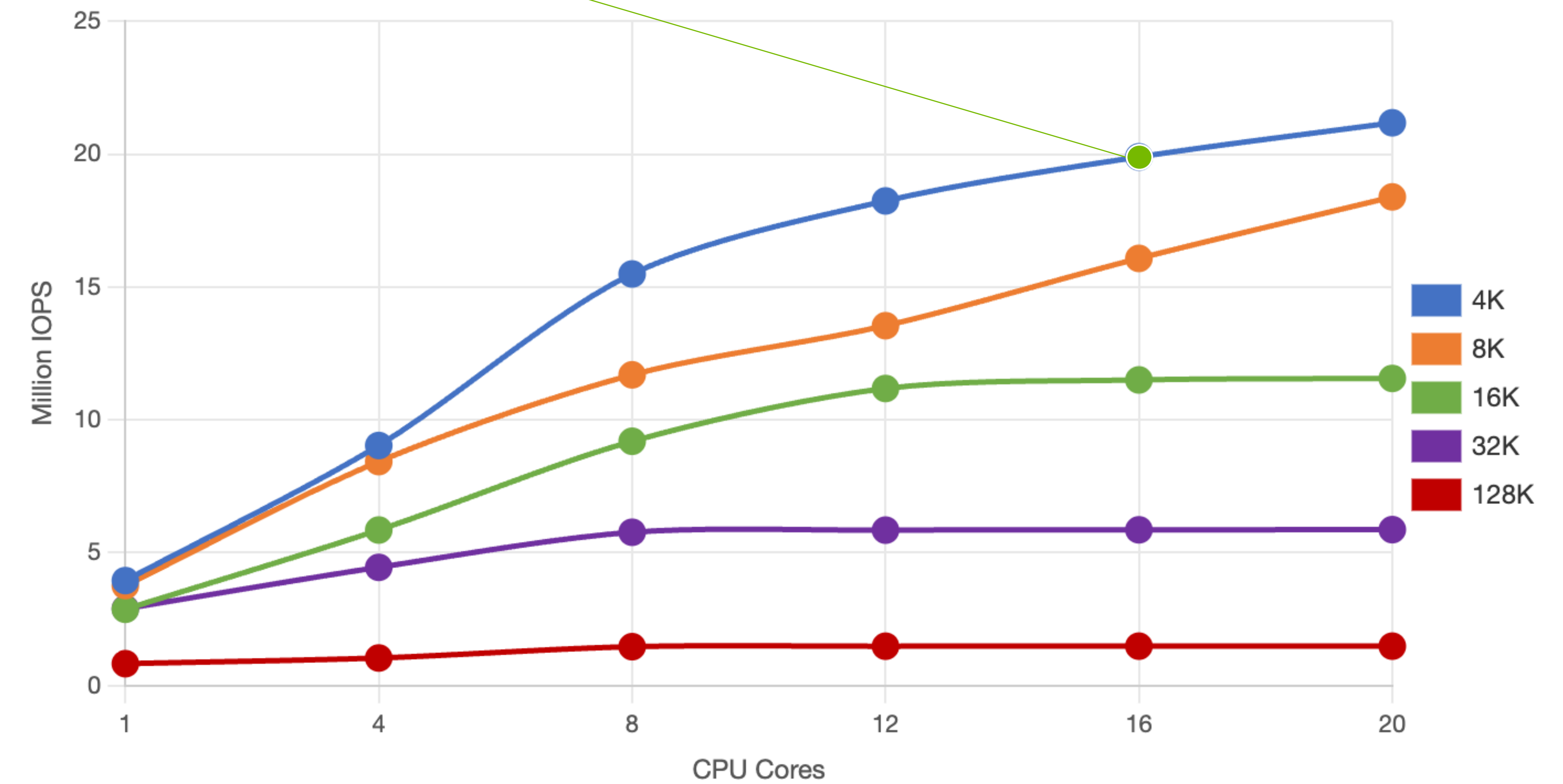
8 Cores for 1.6Tbs

Peak Bandwidth (Gbps) vs CPU Cores



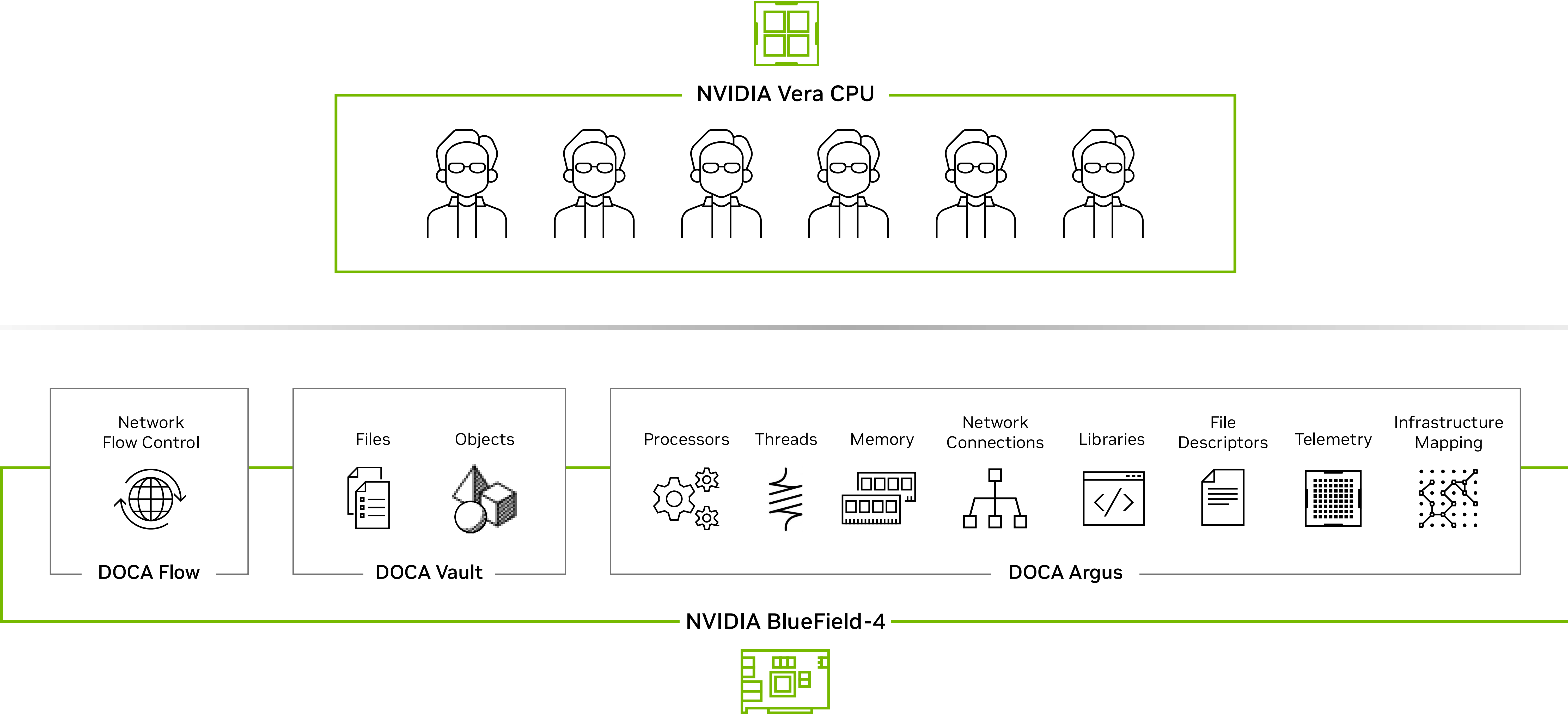
16 Cores for 20M IOPS

Peak IOPS (Millions) vs CPU Cores

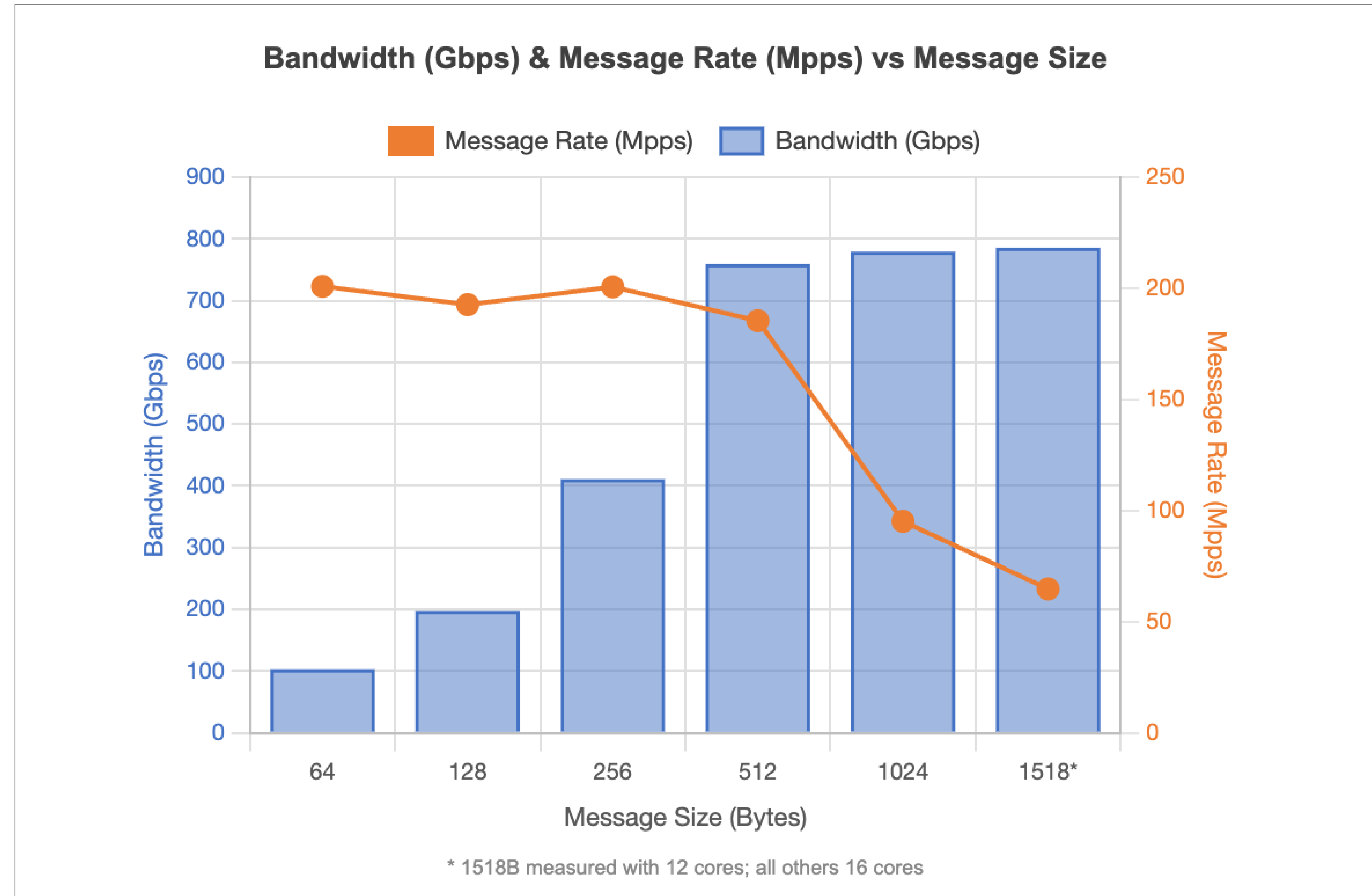


Vera CPU: BF4 Makes Agentic Security Transparent

In-silicon security enforces policy at the speed of AI agents



Line-Rate Networking for the Small Packets AI Infrastructure



Storage-Scale Scales Agent Context With 5x Better Power Efficiency

-  Based on Vera CPU and ConnectX-9
-  3.2Tb/s Storage Access, x10 IOPs, x5 Efficiency
-  Managed by DOCA MEMOS end to end with BlueField-4
-  Co-designed with leading storage partners

GPU HBM	G1 Active KV
System Memory	G2 Staging / Spillover KV
Local Storage	G3 Warm KV reuse
Scale-In Network	G3.5 - CMX
Scale-In Network	G4 Cold or shared KV context



Vera Rubin Co-Designs Every Layer of the AI Factory for Agentic AI

Summary

- **BlueField-4 is the first AI native DPU**
- **Vera Rubin:** 7200 Gb/s DPU with Astra
- **Vera:** Agentic security and confidential compute
- **Scale-In:** End to end service access & security
- **Scale-Out & Across:** Data center services with Astra
- **Storage-Scale:** Efficient AI storage tier

